Terraform challenge

Write a code using Terraform that deploys and achieves the following: -

1. **Create an EC2 Instance on AWS Cloud using the below-mentioned ubuntu AMI and instance_type. Or, (You can use of your choice as well) ami = "ami-0461b1436c9b3c1a3" instance_type = "t2.micro"**

    - Create a folder and open it and created main.tf file, versions.tf

```
main.tf  ×   versions.tf
C:\Users\LENOVO\Desktop\Terraform- Challenge\main.tf r"
provider "aws" {
  region = "us-east-1"
}

resource "aws_instance" "cricbuzz_server" {
  ami           = "ami-0b6d9d3d33ba97d99"
  instance_type = "t3.micro"
  key_name      = "aws"

  tags = {
    Name = "cricbuzz-server"
  }
}
```

```
main.tf  ×   versions.tf  ×
versions.tf > terraform
terraform {
  required_version = ">= 1.0"

  required_providers {
    aws = {
      source  = "hashicorp/aws"
      version = "~> 6.0"
    }
  }
}
```

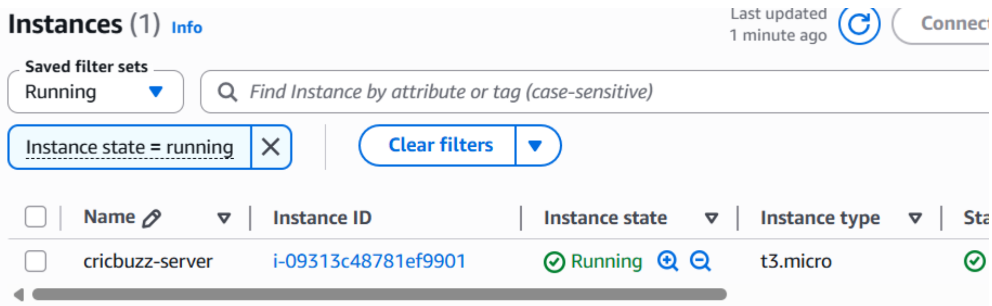
- Now do Terraform init, fmt, validate, plan and apply

```
PS C:\Users\LENOVO\Desktop\Terraform- Challenge> terraform init
Initializing provider plugins found in the configuration...
- Finding hashicorp/aws versions matching "~> 6.0"...
- Installing hashicorp/aws v6.50.0...
- Installed hashicorp/aws v6.50.0 (signed by HashiCorp)

Initializing the backend...


Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!
```

```
PS C:\Users\LENOVO\Desktop\Terraform- Challenge> terraform fmt
main.tf
PS C:\Users\LENOVO\Desktop\Terraform- Challenge> terraform validate
Success! The configuration is valid.

PS C:\Users\LENOVO\Desktop\Terraform- Challenge> terraform plan
```

```
PS C:\Users\LENOVO\Desktop\Terraform- Challenge> terraform apply

Terraform used the selected providers to generate the following execution plan. Re
  + create

Terraform will perform the following actions:

  # aws_instance.cricbuzz_server will be created
  + resource "aws_instance" "cricbuzz_server" {
      + ami                          = "ami-0b6d9d3d33ba97d99"
```

```
Do you want to perform these actions?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.

  Enter a value: yes

aws_instance.cricbuzz_server: Creating...
aws_instance.cricbuzz_server: Still creating... [00m10s elapsed]
aws_instance.cricbuzz_server: Creation complete after 17s [id=i-09313c48781ef9901]

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.
```



2. **Install a web server on the same EC2 Instance. The web server should listen on port 8080.**
   - main.tf file created

```
main.tf    userdata.sh
main.tf > ...
provider "aws" {
  region = "us-east-1"
}

resource "aws_security_group" "web_sg" {
  name = "cricbuzz-web-sg"

  ingress {
    from_port   = 22
    to_port     = 22
    protocol    = "tcp"
    cidr_blocks = ["0.0.0.0/0"]
  }

  ingress {
    from_port   = 8080
    to_port     = 8080
    protocol    = "tcp"
    cidr_blocks = ["0.0.0.0/0"]
  }
```

```
  egress {
    from_port   = 0
    to_port     = 0
    protocol    = "-1"
    cidr_blocks = ["0.0.0.0/0"]
  }
}

resource "aws_instance" "cricbuzz_server" {
  ami           = "ami-0b6d9d3d33ba97d99"
  instance_type = "t3.micro"
  key_name      = "aws"

  vpc_security_group_ids = [aws_security_group.web_sg.id]

  user_data = file("userdata.sh")

  tags = {
    Name = "cricbuzz-server"
  }
}
```

- Userdata.sh file created

```
main.tf    userdata.sh

userdata.sh

apt-get update -y
apt-get install -y nginx

cat > /etc/nginx/sites-available/default << 'EOF'
server {
listen 8080;


location / {
    return 200 "Task 2 Success";
}


}
EOF

systemctl enable nginx
systemctl restart nginx
```

- New do Terraform init, plan and apply

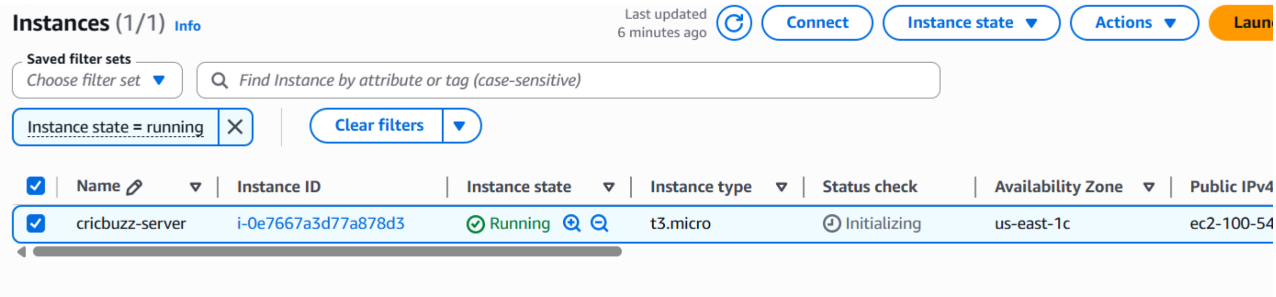
```
PS C:\Users\LENOVO\Desktop\Terraform- Challenge> terraform init
Initializing provider plugins found in the configuration...
- Finding hashicorp/aws versions matching "~> 6.0"...
- Installing hashicorp/aws v6.50.0...
- Installed hashicorp/aws v6.50.0 (signed by HashiCorp)

Initializing the backend...
```

```
PS C:\Users\LENOVO\Desktop\Terraform- Challenge> terraform fmt
main.tf
PS C:\Users\LENOVO\Desktop\Terraform- Challenge> terraform validate
Success! The configuration is valid.

PS C:\Users\LENOVO\Desktop\Terraform- Challenge> terraform plan

Terraform used the selected providers to generate the following execution plan. Resou
  + create

Terraform will perform the following actions:

  # aws_instance.cricbuzz_server will be created
```

```
PS C:\Users\LENOVO\Desktop\Terraform- Challenge> terraform apply

Terraform used the selected providers to generate the following execution plan. R
  + create

Terraform will perform the following actions:

  # aws_instance.cricbuzz_server will be created
  + resource "aws_instance" "cricbuzz_server" {
```

```
  Enter a value: yes

aws_security_group.web_sg: Creating...
aws_security_group.web_sg: Creation complete after 6s [id=sg-07fbf18fae4190fb4]
aws_instance.cricbuzz_server: Creating...
aws_instance.cricbuzz_server: Still creating... [00m10s elapsed]
aws_instance.cricbuzz_server: Creation complete after 16s [id=i-0e7667a3d77a878d3]

Apply complete! Resources: 2 added, 0 changed, 0 destroyed.
```

- Check AWS EC2 instance created



- Connect with SSH to EC2 and check status of Nginx

```
PS C:\Users\LENOVO\Desktop\Terraform- Challenge> ssh -i "C:\Users\LENOVO\Downloads\aws.pem" ubuntu@100.54.106.101
The authenticity of host '100.54.106.101 (100.54.106.101)' can't be established.
ED25519 key fingerprint is SHA256:VUgSV6Bb/n/JvWtaD553IuwAx18F8uvMs+wx1ZM4dHo.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '100.54.106.101' (ED25519) to the list of known hosts.
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1006-aws x86_64)
```

- **Check Status Nginx**

```
ubuntu@ip-172-31-19-7:~$ sudo systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
     Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
     Active: active (running) since Tue 2026-06-16 11:57:00 UTC; 1min 18s ago
 Invocation: 63b23808e6fd4b02a5ba65f9dae5c9ed
       Docs: man:nginx(8)
    Process: 1748 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
    Process: 1750 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
   Main PID: 1751 (nginx)
      Tasks: 3 (limit: 627)
```

- Now check Response

```
ubuntu@ip-172-31-19-7:~$ curl http://localhost:8080
ubuntu@ip-172-31-19-7:~$ curl http://localhost:8080
Task 2 Successubuntu@ip-172-31-19-7:~$
```

3. **Create a Docker-compose file for the project:https://github.com/betawins/cricbuzz-docker-hello-world.gitand point it to port 8000.**

   - Update user-data.sh and created docker-compose file

main.tf | userdata.sh | docker-compose.yml

$ userdata.sh

```bash
#!/bin/bash
set -e

apt-get update -y
apt-get install -y docker.io git golang-go

systemctl enable docker
systemctl start docker

git clone https://github.com/betawins/cricbuzz-docker-hello-world.git /opt/cricbuzz-docker-hello-world

cd /opt/cricbuzz-docker-hello-world

export HOME=/root
export GOPATH=/root/go
export GOMODCACHE=/root/go/pkg/mod
export GOCACHE=/root/.cache/go-build
```

```bash
mkdir -p /root/go/pkg/mod
mkdir -p /root/.cache/go-build

CGO_ENABLED=0 GOOS=linux GOARCH=amd64 go build -a -installsuffix cgo -o hello_world hello_world.go

docker build -t cricbuzz .

docker run -d --name cricbuzz --restart unless-stopped -p 8000:8080 cricbuzz
```

   - **Now do terraform apply**

```
PS C:\Users\LENOVO\Desktop\Terraform- Challenge> terraform apply

Terraform used the selected providers to generate the following execution plan. Resour
  + create

Terraform will perform the following actions:

  # aws_instance.cricbuzz_server will be created
```

- Now connect ssh to Ec2 using public IP

```
PS C:\Users\LENOVO\Desktop\Terraform- Challenge> ssh -i "C:\Users\LENOVO\Downloads\aws.pem" ubuntu@34.229.194.124
The authenticity of host '34.229.194.124 (34.229.194.124)' can't be established.
ED25519 key fingerprint is SHA256:LKcU3An+UGvDGrsaNV8j4QiK0uuwa6cBamuf+Uzem3Q.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '34.229.194.124' (ED25519) to the list of known hosts.
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1006-aws x86_64)
```

- Now check docker status

```
ubuntu@ip-172-31-17-110:~$ sudo systemctl status docker
● docker.service - Docker Application Container Engine
     Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: enabled)
     Active: active (running) since Tue 2026-06-16 13:42:10 UTC; 56s ago
 Invocation: 4922f2a6ea4a440891856dc32276bea6
TriggeredBy: ● docker.socket
       Docs: https://docs.docker.com
   Main PID: 2274 (dockerd)
      Tasks: 9
```

- Now we can see the Docker Compose file and run application on port 800

```
+----[SHA256]-----+
ubuntu@ip-172-31-17-110:~$ curl http://localhost:8000
ubuntu@ip-172-31-17-110:~$ curl http://localhost:8000
Hello, World!ubuntu@ip-172-31-17-110:~$
```

4. **Make the web server point to the above service so that 8080 returns the response from the Docker service.**

- Update userdata.sh file

```
$ userdata.sh
#!/bin/bash
set -e

apt-get update -y
apt-get install -y docker.io git golang-go nginx

systemctl enable docker
systemctl start docker

git clone https://github.com/betawins/cricbuzz-docker-hell

cd /opt/cricbuzz-docker-hello-world

export HOME=/root
export GOPATH=/root/go
export GOMODCACHE=/root/go/pkg/mod
export GOCACHE=/root/.cache/go-build
```

```
mkdir -p /root/go/pkg/mod
mkdir -p /root/.cache/go-build

CGO_ENABLED=0 GOOS=linux GOARCH=amd64 go build -a -installsuffix cgo -o hello_world hello_world.go

docker build -t cricbuzz .

docker run -d --name cricbuzz --restart unless-stopped -p 8000:8080 cricbuzz

cat > /etc/nginx/sites-available/default << 'EOF'
server {
listen 8080;

location / {
    proxy_pass http://127.0.0.1:8000;

    proxy_set_header Host $host;
    proxy_set_header X-Real-IP $remote_addr;
    proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
}
```

- Now do Terraform apply

```
PS C:\Users\LENOVO\Desktop\Terraform- Challenge> terraform apply

Terraform used the selected providers to generate the following execution plan. Resource actions are i
  + create

Terraform will perform the following actions:

  # aws_instance.cricbuzz_server will be created
  + resource "aws_instance" "cricbuzz_server" {
      + ami                              = "ami-0b6d9d3d33ba97d99"
```

- Now connect ssh to Ec2 with EC2 public IP

```
aws_instance.cricbuzz_server: Creating...
aws_instance.cricbuzz_server: Still creating... [00m10s elapsed]
aws_instance.cricbuzz_server: Creation complete after 16s [id=i-054a3e9892f1ebfbb]

Apply complete! Resources: 2 added, 0 changed, 0 destroyed.
PS C:\Users\LENOVO\Desktop\Terraform- Challenge> ssh -i "C:\Users\LENOVO\Downloads\aws.pem" ubuntu@54.235.46.71
The authenticity of host '54.235.46.71 (54.235.46.71)' can't be established.
ED25519 key fingerprint is SHA256:iukBMZVgmOQ3+GewuSiNYcCCEYb41TfRXMJABtT5iww.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '54.235.46.71' (ED25519) to the list of known hosts.
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1006-aws x86_64)
```

- **Configure Nginx reverse proxy so 8080 returns Docker app response**

```
ubuntu@ip-172-31-21-135:~$ curl http://localhost:8080
Hello, World!ubuntu@ip-172-sudo systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
     Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
     Active: active (running) since Tue 2026-06-16 15:58:00 UTC; 4min 54s ago
 Invocation: e642b474533942c5b4c5e77e29058151
       Docs: man:nginx(8)
    Process: 4701 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process on; (code=exit
    Process: 4703 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (code=exited, statu
   Main PID: 4704 (nginx)
      Tasks: 3 (limit: 627)
```

```
Jun 16 15:58:00 ip-172-31-21-135 systemd[1]: Started nginx.service -
ubuntu@ip-172-31-21-135:~$ curl http://localhost:8000
Hello, World!ubuntu@ip-172-31-21-135:~$ curl http://localhost:8080
ubuntu@ip-172-31-21-135:~$
```

5. Configure to Save the Logs in the location /var/log/cb.log

- Update in userdata.sh file nginx server

```
cat > /etc/nginx/sites-available/default << 'EOF'
server {
listen 8080;

access_log /var/log/cb.log;
error_log  /var/log/cb.log;

location / {
    proxy_pass http://127.0.0.1:8000;

    proxy_set_header Host $host;
    proxy_set_header X-Real-IP $remote_addr;
    proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
}

}

EOF

systemctl enable nginx
systemctl restart nginx
```

- Now do Terraform apply and new ec2 created

```
PS C:\Users\LENOVO\Desktop\Terraform- Challenge> terraform apply

Terraform used the selected providers to generate the following execution plan.
  + create

Terraform will perform the following actions:

  # aws_instance.cricbuzz_server will be created
```

- Now connect SSH to EC2 with public Ip

```
aws_instance.cricbuzz_server: Creation complete after 16s [id=i-0b509754a05bebfd0]

Apply complete! Resources: 2 added, 0 changed, 0 destroyed.
PS C:\Users\LENOVO\Desktop\Terraform- Challenge> ssh -i "C:\Users\LENOVO\Downloads\aws.pem" ubuntu@184.72.87.157
The authenticity of host '184.72.87.157 (184.72.87.157)' can't be established.
ED25519 key fingerprint is SHA256:OIRq3YrdiWVL71uTsYybq3X75PMgwG9LKu+u+ciuP4o.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '184.72.87.157' (ED25519) to the list of known hosts.
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1006-aws x86_64)
```

- Nginx access log entries

```
ubuntu@ip-172-31-31-159:~$ curl http://localhost:8080
Hello, World!ubuntu@ip-172-31-31-159:~$ sudo cat /var/log/cb.log
127.0.0.1 - - [16/Jun/2026:16:21:53 +0000] "GET / HTTP/1.1" 200 13 "-" "curl/8.18.0"
ubuntu@ip-172-31-31-159:~$
```

6. Write the Log rotate configuration for the above file

- Update userdata.sh file for log rotate

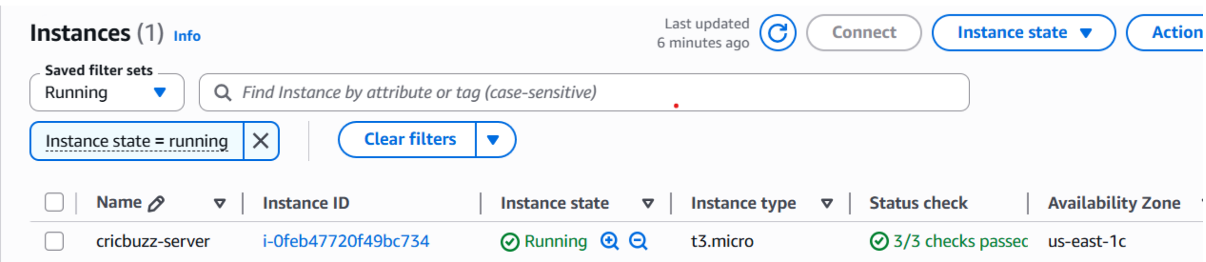
```
# Logrotate configuration

cat > /etc/logrotate.d/cb << 'EOF'
/var/log/cb.log {
daily
rotate 7
compress
missingok
notifempty
create 0644 www-data www-data
}
EOF

systemctl enable nginx
systemctl restart nginx

```

- Now do terraform apply after that new ec2 instance created

```
PS C:\Users\LENOVO\Desktop\Terraform- Challenge> terraform apply

Terraform used the selected providers to generate the following execution plan. Resource actio
  + create

Terraform will perform the following actions:

  # aws_instance.cricbuzz_server will be created
  + resource "aws_instance" "cricbuzz_server" {
      + ami                          = "ami-0b6d9d3d33ba97d99"
```

```
Do you want to perform these actions?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.

  Enter a value: yes

aws_security_group.web_sg: Creating...
aws_security_group.web_sg: Creation complete after 6s [id=sg-0d8369fd89e7f5689]
aws_instance.cricbuzz_server: Creating...
aws_instance.cricbuzz_server: Still creating... [00m10s elapsed]
aws_instance.cricbuzz_server: Creation complete after 16s [id=i-0feb47720f49bc734]

Apply complete! Resources: 2 added, 0 changed, 0 destroyed.
```

- New EC2 instance created



- Now connect SSH to EC2 with public Ip

```
Apply complete! Resources: 2 added, 0 changed, 0 destroyed.
PS C:\Users\LENOVO\Desktop\Terraform- Challenge> ssh -i "C:\Users\LENOVO\Downloads\aws.pem" ubuntu@13.218.35.67
The authenticity of host '13.218.35.67 (13.218.35.67)' can't be established.
ED25519 key fingerprint is SHA256:eDAivRjWB/e+Va0P44Vbdw1tFnLJEJHjNr8+EqFGeFs.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '13.218.35.67' (ED25519) to the list of known hosts.
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1006-aws x86_64)
```

- Now check the all responses

```
ubuntu@ip-172-31-30-5:~$ sudo docker ps
CONTAINER ID   IMAGE     COMMAND          CREATED              STATUS              PORTS                                         NAMES
f1f6a69d2c53   cricbuzz  "/hello_world"   About a minute ago   Up About a minute   0.0.0.0:8000->8080/tcp, [::]:8000->8080/tcp   cricbuzz
ubuntu@ip-172-31-30-5:~$ curl http://localhost:8000
Hello, World!ubuntu@ip-172-31-30-5:~$ curl http://localhost:8080
Hello, World!ubuntu@ip-172-31-30-5:~$ sudo cat /var/log/cb.log
127.0.0.1 - - [16/Jun/2026:17:10:24 +0000] "GET / HTTP/1.1" 200 13 "-" "curl/8.18.0"
ubuntu@ip-172-31-30-5:~$ sudo cat /etc/logrotate.d/cb
/var/log/cb.log {
daily
rotate 7
compress
missingok
notifempty
create 0644 www-data www-data
}
ubuntu@ip-172-31-30-5:~$
```

- Now destroy all

```
ubuntu@ip-172-31-30-5:~$ exit
logout
Connection to 13.218.35.67 closed.
PS C:\Users\LENOVO\Desktop\Terraform- Challenge> terraform destroy
aws_security_group.web_sg: Refreshing state... [id=sg-0d8369fd89e7f5689]
aws_instance.cricbuzz_server: Refreshing state... [id=i-0feb47720f49bc734]

Terraform used the selected providers to generate the following execution plan. Resource actions are i
  - destroy

Terraform will perform the following actions:

  # aws_instance.cricbuzz_server will be destroyed
```

```
  There is no undo. Only 'yes' will be accepted to confirm.

  Enter a value: yes

aws_instance.cricbuzz_server: Destroying... [id=i-0feb47720f49bc734]
aws_instance.cricbuzz_server: Still destroying... [id=i-0feb47720f49bc734, 00m10s elapsed]
aws_instance.cricbuzz_server: Still destroying... [id=i-0feb47720f49bc734, 00m20s elapsed]
aws_instance.cricbuzz_server: Still destroying... [id=i-0feb47720f49bc734, 00m30s elapsed]
aws_instance.cricbuzz_server: Destruction complete after 32s
aws_security_group.web_sg: Destroying... [id=sg-0d8369fd89e7f5689]
aws_security_group.web_sg: Destruction complete after 2s

Destroy complete! Resources: 2 destroyed.
PS C:\Users\LENOVO\Desktop\Terraform- Challenge>
```